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Keywords: NLP; Multilingual; Multimodal; LLMs.

Summary

Research Scientist of Multimodal NLP team at Creative AI Lab, Sony. Ph.D. specializing in NLP, focusing on **Multilingual and Multimodal LLM**. These works have been presented at conferences such as **ACL**, **EMNLP**, **ICLR**, **NAACL**, **ICML**, **TMLR**, **VLDB**, **EACL**, **AAAI**, etc. Served as area chair and reviewer in above top conferences and organized workshops at NeurIPS. Received JSPS DC fellowship.

Research Experiences

- Research Scientist**, *Sony Group Corporation* 2024.10 – Present
- Leading a small research team of multimodal LLM: formulating novel research problems from ambiguity; driving academic impact with universities and production deployment with engineers.
 - Research interests: multimodal LLMs for understanding, retrieving and benchmarking contents in language, vision and music, etc., for potentially enhancing various business scenarios. [M1, M2, M3, M4, M6, W2, W4, C12, C13, C14, C17].
- Ph.D. Researcher**, *Tsuruoka Lab at UTokyo* 2021.10 – 2024.09
- Word-level preference for LLM-based machine translation [C10]; Word alignment pre-training [C5, C11];
 - Text embeddings, generation and compression [M5]; Cross-lingual text embeddings [C8, C7].
- Visiting Student**, *UC San Diego* 2023.01 – 2023.02
- Generalized diffusion model [C9].
- Research Intern**, *Microsoft Research Asia* 2020.04 – 2022.02
- Multiple positive examples in sentence embeddings with contrastive learning [C4], work done at STCA.
 - Acceleration of language pre-training via better rare word representation [C3], work done at MSR Asia.
- R&D Intern**, *Baidu Research* 2019.10 – 2020.02
- Multi-task transportation prediction [C2]; Route mode recommendation in Baidu Maps.

Education

- The University of Tokyo**, Ph.D. Degree in Information Science & Technology. 2021.10 – 2024.09
- Ph.D. at Tsuruoka Lab; JSPS DC2 Fellowship; JST SPRING Fellowship.
- Peking University**, M.Sc. Degree in Computer Software & Theory. 2018.09 – 2021.07
- Sichuan University**, B.Eng. Degree in Computer Science & Technology. 2014.09 – 2018.07

Advising and Services

- Advised Students**: Youngtaek Oh [M3], Daeyong Kwon [M6], Dominick Reilly [M4], Jongsuk Kim [M1], Zhongtao Miao [M5, C8, C11], Kaiyan Zhao [C7, W3], Yutao Xie [J1].
- Area Chair of ACL Rolling Review (2025 – Present); NeurIPS (2026 – Present).**
- Reviewer of ACL Rolling Review, EMNLP, ICLR, NeurIPS, ICML, etc.** 2021 – Present
- Organizer of “NeurIPS 2025 GenProCC Workshop”.** 2025
- DC2 Fellowship of JSPS.** 2024
- Teaching Assistant of “Social Implications of Computing” at Peking University.** 2019

Publications

‡ indicates corresponding author; * indicates equal contribution.

[M6] Daeyong Kwon, Qiyu Wu[‡], Shinobu Kuriya, Junghyun Koo, Shuyang Cui, Zhi Zhong, Wei-Hsiang Liao, Hiromi Wakaki, Yuki Mitsufuji. “MusT-BENCH: Benchmarking Music Temporal Intelligence”, *manuscript*.

[M5] Zhongtao Miao, Qiyu Wu, Yoshimasa Tsuruoka. “GRC: Unifying Reasoning-Driven Generation, Retrieval and Compression”, *manuscript*.

[M4] Dominick Reilly, Qiyu Wu[‡], Hiromi Wakaki, Srijan Das, Yuki Mitsufuji. “Mixture of Probes: Learning with Privileged Modalities in Multimodal LLMs Through Probing”, *manuscript*.

[M3] Youngtaek Oh, Qiyu Wu[‡], Hiromi Wakaki, Junmo Kim, Yuki Mitsufuji. “Syn-Omni: Structured Specialization and Progressive Collaboration for Omnimodal Embeddings”, *manuscript*.

[M2] Qiyu Wu, Shuyang Cui, Satoshi Hayakawa, Wei-Yao Wang, Hiromi Wakaki, Yuki Mitsufuji. “MCA: Modality Composition Awareness for Robust Composed Multimodal Retrieval”, *manuscript*.

[M1] Jongsuk Kim, Qiyu Wu[‡], Zhuoyuan Mao, Hiromi Wakaki, Junmo Kim, Yuki Mitsufuji. “MLLMCLIP: Feature-Level Distillation of MLLM for Robust Vision-Language Representations”, *manuscript*.

[C17] Silin Gao, Hao Zhao, Zeming Chen, Sepideh Mamooler, Antara Raaghavi Bhattacharya, Qiyu Wu, Hiromi Wakaki, Yuki Mitsufuji, Li Mi, Syrielle Montariol, Antoine Bosselut. “DynaView: Schema-Guided World Modeling for Understanding Hierarchical Visual Dynamics”, (*ICML 2026*).

[C16] Shaomu Tan, Ryosuke Mitani, Ritvik Choudhary, Qiyu Wu, Toshiyuki Sekiya, Christof Monz. “Remedy-R: Generative Reasoning for Machine Translation Evaluation without Error Annotations”, (*ACL 2026*).

[C15] Shuyang Cui, Zhi Zhong, Qiyu Wu, Zachary Novack, et al. “BREAK-THE-BEAT! Controllable MIDI-to-Drum Audio Synthesis”, (*ICASSP 2026*).

[C14] Seunghoon Doh, Junghyun Koo, Marco A Martínez-Ramírez, Woosung Choi, Wei-Hsiang Liao, Qiyu Wu, Juhan Nam, Yuki Mitsufuji. “LLM2Fx-Tools: Tool Calling For Music Post-Production”, (*ICLR 2026*).

[C13] Wei-Yao Wang, Kazuya Tateishi, Qiyu Wu, Shusuke Takahashi, Yuki Mitsufuji. “VIRTUE: Visual-Interactive Text-Image Universal Embedder”, (*ICLR 2026*).

[C12] Zhuoyuan Mao, Mengjie Zhao, Qiyu Wu, Hiromi Wakaki, Yuki Mitsufuji. “DeepResonance: Enhancing Multimodal Music Understanding via Music-centric Multi-way Instruction Tuning”, (*EMNLP 2025 Main*).

[C11] Zhongtao Miao, Qiyu Wu, Masaaki Nagata, and Yoshimasa Tsuruoka. “Improving Word Alignment Using Semi-Supervised Learning”, (*ACL 2025 Findings*).

[W4] Zhuoyuan Mao, Mengjie Zhao, Qiyu Wu, Zhi Zhong, Wei-Hsiang Liao, Hiromi Wakaki, Yuki Mitsufuji. “Cross-Modal Learning for Music-to-Music-Video Description Generation”, in Proceedings of the 10th Workshop on Representation Learning for NLP (*RePL4NLP-2025*).

[W3] Kaiyan Zhao, Qiyu Wu, Zhongtao Miao, Yoshimasa Tsuruoka. “Prompt Tuning Can Simply Adapt Large Language Models to Text Encoders”, in Proceedings of the 10th Workshop on Representation Learning for NLP (*RePL4NLP-2025*).

[J2] Yuting Tang, Xin-Qiang Cai, Yao-Xiang Ding, Qiyu Wu, Guoqing Liu, Masashi Sugiyama. “Reinforcement Learning from Bagged Reward”, in Transactions on Machine Learning Research (*TMLR; ICML 2026 Poster Track*).

[W2] Qiyu Wu, Mengjie Zhao, Yutong He, Lang Huang, Junya Ono, Hiromi Wakaki, Yuki Mitsufuji. “Towards Reporting Bias in Visual-language Datasets: Bi-modal Data Augmentation by Decoupling Object-attribute Association”, (*ICCV 2025 Workshop Multimodal Representation and Retrieval*).

[C10] Qiyu Wu, Masaaki Nagata, Zhongtao Miao, Yoshimasa Tsuruoka. “Word Alignment as Preference for Machine Translation”, (*EMNLP 2024 Main, Oral*).

[W1] Qiyu Wu, Zilong Wu, Yoshimasa Tsuruoka. “SGA: Scene Graph Alignment for Evaluation of Text-to-Image Generation”, (*KDD 2024 Workshop HI-AI*).

- [C9] Guangyi Liu, Yu Wang, Zeyu Feng, Qiyu Wu, Liping Tang, Yuan Gao, Zhen Li, Shuguang Cui, Julian McAuley, Eric P. Xing, Zichao Yang, Zhiting Hu. “Unified Generation, Reconstruction, and Representation: Generalized Diffusion with Adaptive Latent Encoding-Decoding”, in *the Forty-first International Conference on Machine Learning (ICML 2024)*.
- [C8] Zhongtao Miao, Qiyu Wu, Kaiyan Zhao, Zilong Wu, Yoshimasa Tsuruoka. “Enhancing Cross-lingual Sentence Embedding for Low-resource Languages with Word Alignment”, in *the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024, findings)*.
- [C7] Kaiyan Zhao*, Qiyu Wu*, Xin-Qiang Cai, Yoshimasa Tsuruoka. “Leveraging Multi-lingual Positive Instances in Contrastive Learning to Improve Sentence Embedding”, in *the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2024, main conference long paper)*.
- [C6] Shiwen Wu, Qiyu Wu, Honghua Dong, Wen Hua, Xiaofang Zhou. “Blocker and Matcher Can Mutually Benefit: A Co-Learning Framework for Low-Resource Entity Resolution.”, *Proc. VLDB Endow. (2024). Appear at Very Large Databases Conf. (VLDB 2024, research track paper)*.
- [C5] Qiyu Wu, Masaaki Nagata, Yoshimasa Tsuruoka. “WSPAlign: Word Alignment Pre-training via Large-Scale Weakly Supervised Span Prediction”, in *the 61st Annual Meeting of the Association for Computational Linguistics (ACL 2023 Main)*.
- [C4] Qiyu Wu, Chongyang Tao, Tao Shen, Can Xu, Xiubo Geng, Daxin Jiang. “PCL: Peer-Contrastive Learning with Diverse Augmentations for Sentence Embeddings”, in *Conference on Empirical Methods in Natural Language Processing (EMNLP 2022 Main)*.
- [J1] Yutao Xie, Qiyu Wu, Wei Chen and Tengjiao Wang. “Stable Contrastive Learning for Self-Supervised Sentence Embeddings With Pseudo-Siamese Mutual Learning”, in *IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)*, vol. 30, pp. 3046-3059, 2022.
- [C3] Qiyu Wu*, Chen Xing*, Yatao Li, Guolin Ke, Di He, Tie-Yan Liu. “Taking Notes on the Fly Helps Language Pre-Training”, in *the Ninth International Conference on Learning Representations (ICLR 2021)*.
- [C2] Hao Liu*, Qiyu Wu*, Fuzhen Zhuang, Xinjiang Lu, Dejing Dou, Hui Xiong. “Community-Aware Multi-Task Transportation Demand Prediction”, in *Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI 2021)*.
- [C1] Dechuan Yang*, Qiyu Wu*, Wei Chen, Tengjiao Wang, Zhen Qiu, Di Liu. “Stance Detection with Stance-wise Convolution Network”, in *the 2022 CCF International Conference on Natural Language Processing and Chinese Computing*.

[J]: Journal paper; [C]: Conference paper; [W]: Workshop paper; [M]: Manuscript.